PRJ301: Java Web Application Development

Slot 1, 2, 3: Introduction, Servlet, and Web Application

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Objectives

- **Introduction to Java Web Application:**
  - The core and basic of Java web server technologies.
  - Web design vs server technologies.
- **Setup Environment:**
  - JDK 24 or higher.
  - Servlet container: Tomcat 10.1.
  - Integrated Development Environment: NetBeans 25.
- **Creating/Building the first application:**
  - Learn to create Servlet.
  - Learn to create JSP.
  - Deploy the web application.
  - **AI Integration:** Effective Prompt Engineering for Java Web.

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Outline

Slot 1: Web Foundations & Environment

Slot 2: Servlet Lifecycle & Request Handling

Slot 3: HTML Forms, App Structure & Practice

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Slot 1: Web Foundations & Environment

What is HTML?

- HTML stands for **Hyper Text Markup Language**.
- It describes the structure of web pages using **tags**.
- HTML tags are keywords surrounded by angle brackets like `<html>`.
- Tags normally come in pairs: start tag (`<b>`) and end tag (`</b>`).
- Browsers read HTML documents and display them as web pages (interpreting tags rather than displaying them).

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HTML Document Example

```
1 <html>
2     <body>
3         <h1>My First Heading</h1>
4         <p>My first paragraph</p>
5         <a href="http://cms.fpt.edu.vn">Studying place</a>
6     </body>
7 </html>
```

Listing 1: A basic web page

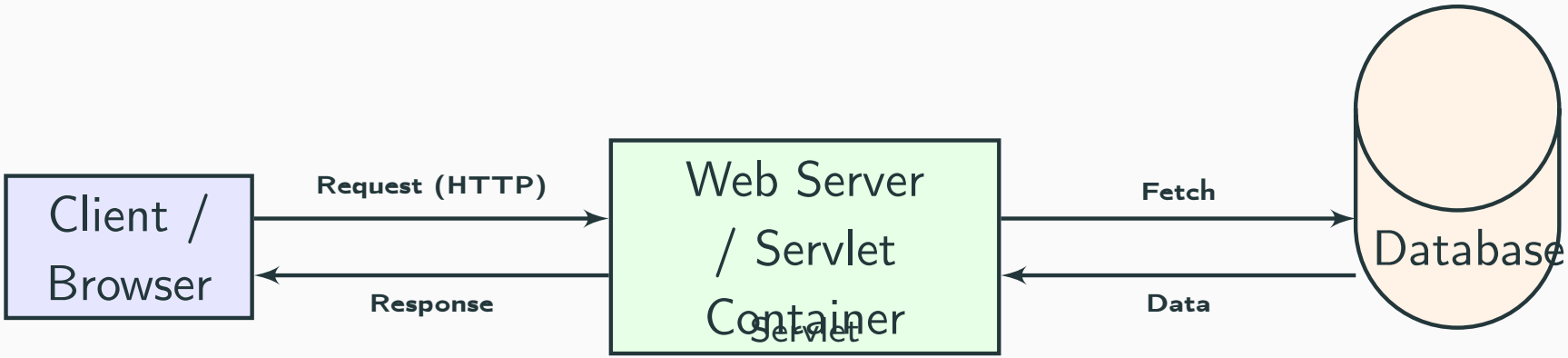
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What is a Servlet?

- **Servlets** are small Java programs that run on a web server.
- They build **dynamic** web pages (content changes per request).
- They receive and respond to requests from web clients via **HTTP**.
- Java Servlet technology provides a portable way to deliver user-oriented content.

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Java Web Architecture



- **jakarta.servlet** package provides core interfaces.
- **ServletRequest:** Encapsulates communication from client to server.
- **ServletResponse:** Encapsulates communication from server to client.

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HTTP Requests

An HTTP Request consists of: Method, Request URL, Headers, and Body.

- **GET**: Retrieves resource (idempotent and safe).
- **POST**: Sends data of unlimited length.
- **HEAD**: Returns headers only.
- **PUT/DELETE**: Stores/Removes a resource.

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HTTP Responses

An HTTP Response contains: Result code, Headers, and Body.

Common Status Codes

- **200 OK**: Everything is fine.
- **404 Not Found**: Resource missing.
- **401 Unauthorized**: Authentication required.
- **500 Internal Server Error**: Server-side failure.

Reference: `https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Status`

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HTTP Response Structure

A typical HTTP response from the server consists of:

1. **Status Line**: HTTP version, status code, and message.
2. **Response Headers**: Metadata about the response.
3. **Blank Line**: Separates headers from the body.
4. **Response Body**: The actual content (HTML, JSON, etc.).

Example Response

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 45

<html><body><h1>Hello World!</h1></body></html>
```

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ServletResponse & HttpServletResponse

- **ServletResponse**: The base interface for sending data to the client.
- **HttpServletResponse**: Extends `ServletResponse` with HTTP-specific functionality.

Common Tasks

- `setContentType(String type)`: Sets the MIME type (e.g., `text/html`).
- `getWriter()`: Returns a `PrintWriter` for character data.
- `getOutputStream()`: For binary data.
- `sendRedirect(String url)`: Redirects the client to another URL.
- `addCookie(Cookie c)`: Sends a cookie to the browser.

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Slot 1 Review: Open-Ended Questions

1. How does a dynamic web application differ from a static one in terms of server-side processing?
2. Explain the journey of an HTTP Request from the moment a user clicks a link until they see a response.
3. In a Client-Server model, why is it considered "stateless"? How might this affect web application design?
4. Compare the structure of an HTTP Request and an HTTP Response. What key information is unique to each?

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Slot 1 Review: Discussion Questions

1. How is a dynamic web application different from a static one in server-side processing?
2. Explain the path of an HTTP Request from a click to a response.
3. In the Client-Server model, why is it called "stateless"? How does this affect design?
4. Compare the structure of an HTTP Request and Response. What info is unique to each?

(Translated to Vietnamese for context in the source file:)

1. Ung dung web dong khac tinh nhu the nao?
2. Hanh trinh cua HTTP Request?
3. Tai sao goi la stateless?
4. So sanh Request va Response?

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Slot 2: Servlet Lifecycle & Request Handling

A Simple Servlet (Jakarta EE 10+)

```java
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;

@WebServlet(name = "FirstServlet", urlPatterns = {"/FirstServlet"})
public class FirstServlet extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest request,
     HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<h1>First Servlet</h1>");
    }
}
```

Listing 2: FirstServlet.java

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The HttpServlet Class

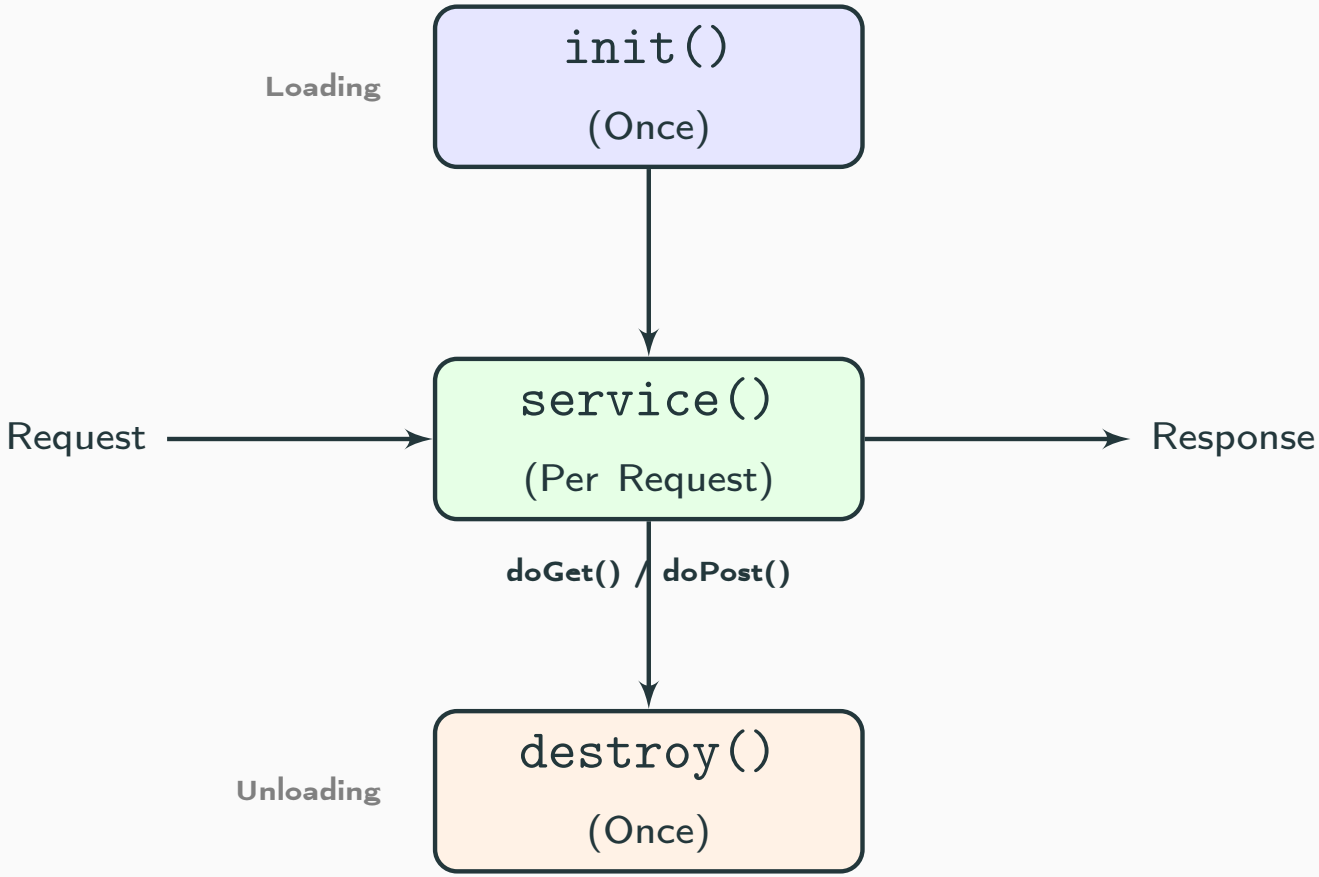
- `HttpServlet` is an abstract class that simplifies handling HTTP requests.
- It provides specialized methods for each HTTP verb:

Method	Description
doGet	**Handles GET requests (default method).**
doPost	**Handles POST requests (for sending data).**
`doHead`	Returns headers only (no body).
`doPut`	Handles PUT requests (updates/replaces).
`doDelete`	Handles DELETE requests.
`doOptions`	Returns supported HTTP methods.
`doTrace`	Returns the request for debugging.

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The Servlet Life Cycle

The container manages the servlet through three main stages:



- **init()**: Initialize resources (DB connections, config).
- **service()**: Dispatches to `doGet`, `doPost`, etc.
- **destroy()**: Clean up resources before removal.

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Request Headers

Browsers send **metadata** via headers:

- `Accept`, `Accept-Charset`, `Cookie`, `Host`, `Referer`, `User-Agent`.

Accessing Headers in Java

`request.getHeader("User-Agent");`

`request.getHeaderNames();` (returns `Enumeration<String>`)

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Example: Displaying Request Headers

```
1  public void doGet(HttpServletRequest request, HttpServletResponse
       response)
2          throws ServletException, IOException {
3      response.setContentType("text/html");
4      PrintWriter out = response.getWriter();
5      Enumeration<String> names = request.getHeaderNames();
6      out.println("<table border='1'>");
7      while(names.hasMoreElements()) {
8          String name = names.nextElement();
9          out.println("<tr><td>" + name + "</td><td>"
10             + request.getHeader(name) + "</td></tr>");
11     }
12     out.println("</table>");
13 }
```

Listing 3: ShowRequestHeaders.java

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Slot 3: HTML Forms, App Structure & Practice

HTML Forms: User Input

- A **Form** is an area that contains various input elements.
- **Form Elements** allow users to enter information:
  - Text fields / Textarea
  - Drop-down menus (Select)
  - Radio buttons
  - Checkboxes
  - Submit / Reset buttons

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HTML Forms

Forms collect user input and send it to the server.

```
<form action="LoginServlet" method="POST">
    User: <input type="text" name="user"/><br/>
    Pass: <input type="password" name="pass"/><br/>
    <input type="submit" value="Login"/>
</form>
```

Listing 4: Login Form

- **action**: Target URL.
- **method**: GET (parameters in URL) or POST (parameters in body).

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Handling Form Data in Servlet

Call getParameter method of the HttpServletRequest, supplying the parameter name as an argument

```
1  public void doPost(HttpServletRequest request, HttpServletResponse
       response)
2          throws ServletException, IOException {
3      String u = request.getParameter("user");
4      String p = request.getParameter("pass");
5
6      response.setContentType("text/html");
7      PrintWriter out = response.getWriter();
8      out.println("<html><body>");
9      out.println("<h1>You sent me:</h1>");
10     out.println(u + "<br>" + p);
11     out.println("</body></html>");
12 }
```

Listing 5: doPost Method

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Discussion: GET vs. POST Methods

- **Question 1:** If you type a URL directly into the browser address bar (e.g., `localhost:8080/App/MyServlet`), which method is used?
- **Question 2:** Why does `HttpServlet` have both `doGet()` and `doPost()`? Can one Servlet handle both?

Key Differences to Remember

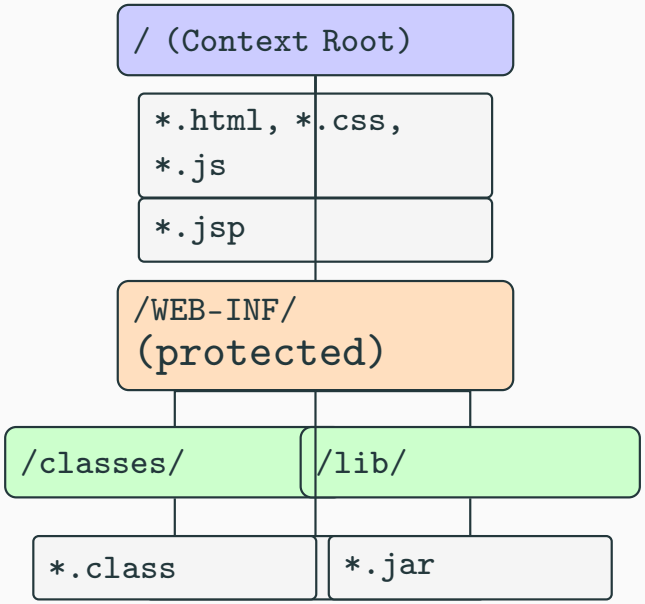
- **GET**: Parameters are visible in the URL. Limited data size. Used for "fetching" data. *Triggered by: Links, Address Bar, Form (method=GET).*
- **POST**: Parameters are hidden in the Request Body. Large data size. Used for "submitting/updating" data. *Triggered by: Form (method=POST) only.*

Common Confusion

Browsers **always** use GET when you hit Enter in the address bar. You cannot send a POST request just by typing a URL!

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Web Application Structure



Key Rules

- Files in / are publicly accessible via URL.
- **/WEB-INF/** is never served directly to clients — safe to store config and code.
- **web.xml** is the Deployment Descriptor (Servlet mappings, welcome files, etc.).
- **/lib/** holds third-party JARs (e.g., JDBC drivers).

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Packaging & Deployment

- **WAR (Web ARchive)**: A single file containing the entire web application.
- Tomcat automatically deploys WAR files in the `webapps` folder.
- **Welcome Files**: Defined in `web.xml`, specifies which file to show if no filename is requested (e.g., `index.html`).

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Classwork 1: Welcome App

Task
Print welcome message using parameters `name` and `color` from URL.

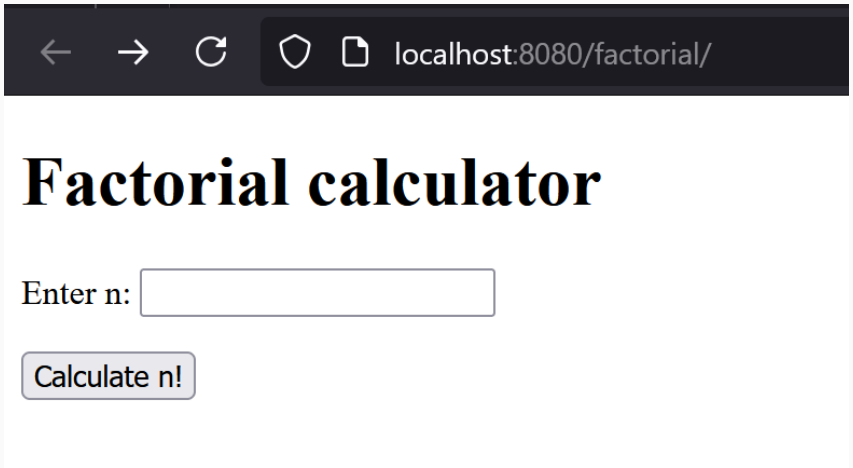
E.g.: http://localhost:8080/first-app/welcome?name=John&color=blue

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Classwork 2: Factorial Calculator

Task
Calculate $n!$ for $1 < n < 21$. Handle invalid inputs.

- URL pattern: `/factorial`
- Constraint: $1 < n < 21$
- Input: integer n



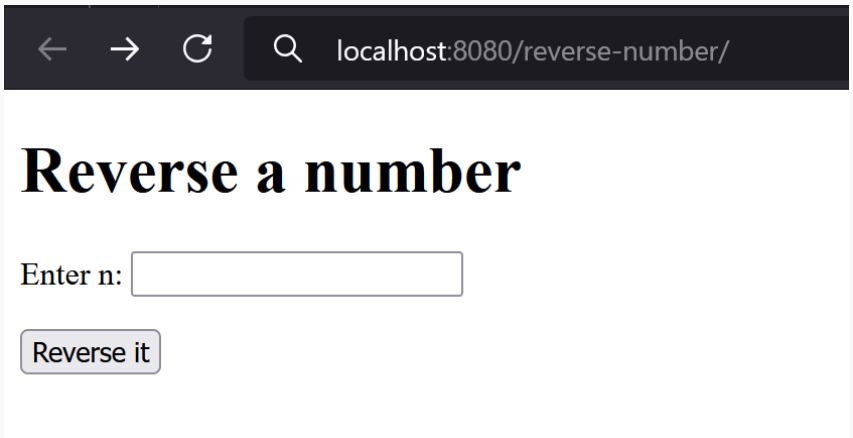
Legacy Reference UI

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Classwork 3: Number Reversal

Task
Reverse a 3-digit number
($99 < n < 1000$).

- URL pattern: `/reverse-num`
- Input: integer n
- If entered n is not a 3-digit number, output an error.



Legacy Reference UI

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Summary: Slot 1, 2, 3

- **Web Foundations:** HTTP Protocol (Stateless, Request/Response), HTML basics.
- **Servlet Basics:** Dynamic content, `jakarta.servlet` API, and `HttpServlet` class.
- **Servlet Lifecycle:** `init()`, `service()` (doGet/doPost), and `destroy()`.
- **Request Handling:** Accessing parameters (`getParameter`) and headers (`getHeader`).
- **App Structure:** Context Root, `WEB-INF`, `web.xml`, and WAR packaging.

Next Steps
Practice with the Classwork tasks and prepare for **JSP & JDBC** in the next slots.

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Questions?

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